



Arora V Successive Approximation Register Analog to Digital Converter (SAR ADC)

User Guide

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Revision History

Date	Version	Description
1/20/2025	1.0E	Initial version published.
6/27/2025	1.0.1	• 3.2 Input Channel Selection (Voltage Divider Ratio) added. • SAR ADC instantiation details added.

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1 About This Guide

1.1 Purpose

Arora V Analog to Digital Converter (SAR ADC) User Guide is to help you quickly learn the features and usage of Arora V SAR ADC by introducing to the functions, ports, configuration, etc.

1.2 Related Documents

The latest user guides are available on the GOWINSEMI website. You can find the related documents at www.gowinsemi.com:

- [DS981, GW5AT series of FPGA Products Data Sheet](#)
- [DS1103, GW5A series of FPGA Products Data Sheet](#)
- [SUG100, Gowin Software User Guide](#)

1.3 Terminology and Abbreviations

The terminology and abbreviations used in this manual are as shown in [Table 1-1](#).

Table 1-1 Terminology and Abbreviations

Terminology and Abbreviations	Full Name
ADC	Analog to Digital Converter
CIC Filter	Cascaded Integrator–comb Filter
FPGA	Field Programmable Gate Array
IP	Intellectual Property
OSC	Oscillator
SAR	Successive Approximation Register
SRAM	Static Random Access Memory

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 Overview

Arora V FPGA 60K products integrate 13-bit SAR ADC designed for low power and low-leakage current. With the FPGA's programmable logic and integrated voltage references, the ADC is well-suited for meeting internal analog-to-digital conversion requirements. At the same time, the FPGA provides flexible and configurable GPIO interfaces and ADC analog differential signal interfaces to connect to the voltage channel of the ADC, which can meet the external voltage data collection and monitoring requirements.

2.1 Features

The main features of Arora V SAR ADC are as follows:

- Number of ADCs: 1
- Number of ADC input channels: up to 122^{[1][2]} channels supported.
- Reference voltage source: Built-in
- Bit width accuracy: 13 bits
- Max. sampling rate: 10Msps
- Sampling Clock: 10MHz~ 300MHz
- Single-ended input signal range: 0V -1V
- Differential signal range: -1V ~ 1V
- Voltage sensor accuracy: +/-0.3mV

Note!

- ^[1] Please refer to the relevant pinout manual for details on the supported number of input channels.
- ^[2] All Bank differential GPIOs that support bus connections for ADC_INPUT are available for usage; it is recommended to use GPIOs from Bank 2.

2.2 Functional Description

2.2.1 Overview

The Arora V SAR ADC integrates an analog SAR modulator for analog-to-digital conversion and provides differential GPIO input interfaces to support external voltage signal acquisition. Both differential and single-ended signal inputs are supported.

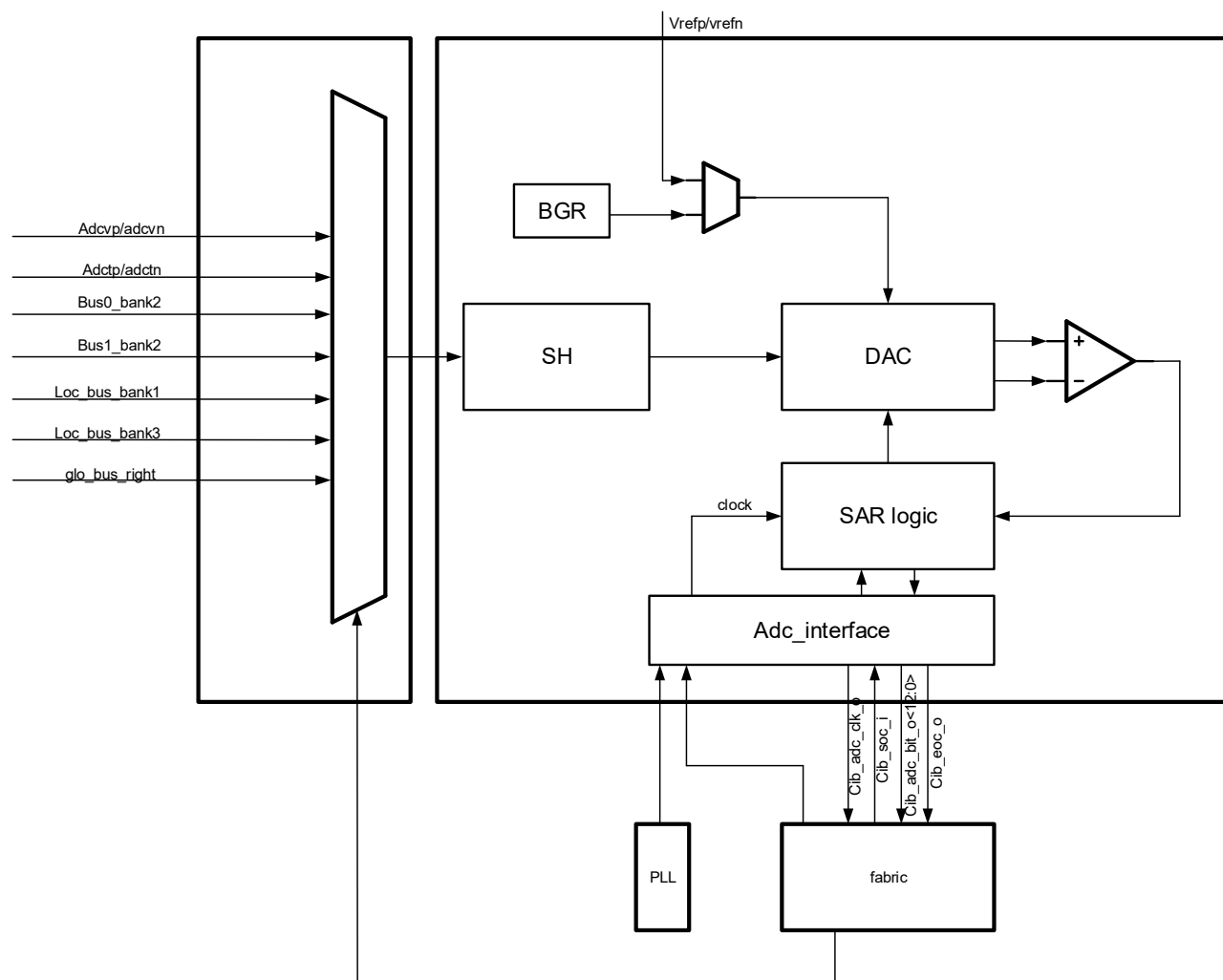
Arora V SAR ADC integrates a high-accuracy internal reference and supports external reference sources, with its supply voltage monitoring offering low-power and high-precision performance.

The Arora V SAR ADC integrates an internal voltage signal processing module, eliminating the need for an external reference voltage source. This enables accurate voltage measurements while reducing system cost for the user.

2.2.2 Architecture

[Figure 2-1](#) shows the structure diagram of GW5A-60/GW5AT-60 SAR ADC.

Figure 2-1 60K SAR ADC Diagram



The GW5A-60/GW5AT-60 SAR ADC supports on-chip voltage monitoring mode. By configuring control signals, users can select different measurement paths to monitor the power supply voltages of internal FPGA IP modules, I/O Banks, core voltage, SRAM voltage, etc. External voltage signals can be routed to the ADC for quantization through the differential GPIO pins of Bank 0/1/2/3/4/5/6/7 or through dedicated differential ADC input pins.

The GW5A-60/GW5AT-60 SAR ADC offers the option to select either the user logic clock (CLK) or a PLL input as its clock source, allowing designers to balance power efficiency and performance based on system requirements.

2.2.3 SAR ADC Conversion Timing

The SAR ADC requires N sampling clock cycles to convert the analog input signal and generate the corresponding digital output. When the rising edge of the soc signal comes, and the ADC enable signal `adceni` is enabled (active-high), it will trigger ADC to perform a

single sampling operation. Upon completion, the eoc signal is driven high to indicate the end of sampling, and the sampled data is output on adcbitt[12:0].

adcbitt[12:0] is a 13-bit unsigned value. The actual measured voltage (in volts) can be calculated using the following formula: $(\text{adcbitt}[12:0]/4096-1) \times V_{\text{ref}} \times \text{Divider Ratio}$. The unit is volts (V).

Figure 2-2 SAR ADC Conversion Timing

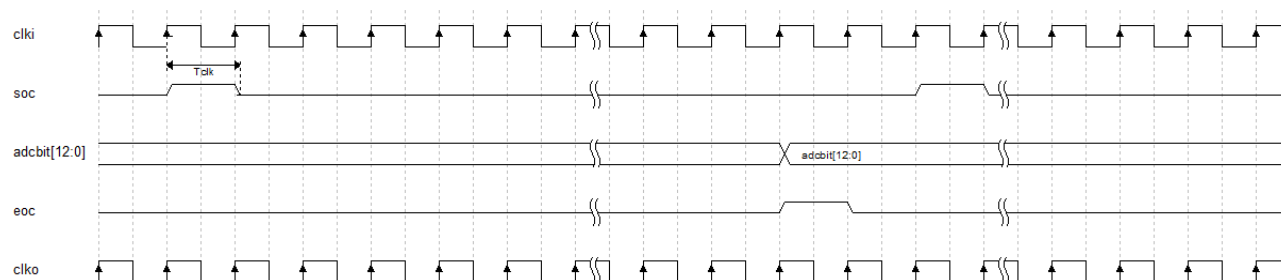


Table 2-1 SAR ADC Timing Parameters

Symbol	Description	Spec.		Unit
		Min.	Max.	
T _{clk}	Clock cycle	3.125	100	ns
T _S	soc setup time	0.8	-	ns
T _H	soc hold-up time	0.8	-	ns
N _{conversionCycle}	Conversion cycles	22	32	Cycle

2.2.4 Electrical Characteristic Parameters

Table 2-2 SAR ADC Electrical Parameters

Parameter	Description	Spec.			Unit
		Min.	Typ.	Max.	
DC precision					
Output	Digital output bits	-	13	-	Bit
INL	Integral nonlinearity	-2	-	2	LSB
DNL	Differential nonlinearity	-1	-	1	LSB
Analog Input					
Vref	External Input Voltage Reference	-	1.25	-	V
CH[7: 0]	Single-ended input range	0	-	1	V
CIN	Input capacitance ^[1]	-	1	-	pF
Slew Rate					
Fs	Ratio	-	-	10	MHz

Parameter	Description	Spec.			Unit
		Min.	Typ.	Max.	
CLK	Master Clock	10	-	300	MHz
N _{ConversionCycles}	Conversion Cycles	22	-	32	Clock cycle
Dynamic Characteristic Parameters					
SNRFS ^[2]	Signal Noise Ratio	-	60	-	dB
ENOB	Valid output data bits	-	9.6	-	Bit
V _{INBW}	Dedicated Input Pin -3dB Bandwidth	-	160	-	MHz
Supply voltage ^[3]					
V _{dd_a}	Analog core voltage	-	0.9	-	V
V _{dd_dig}	Digital voltage	-	0.9	-	V
V _{ddx}	Analog voltage	-	1.8	-	V

Note!

- ^[1] Input capacitance through the dedicated pin.
- ^[2] Test conditions: Vin= -0.5dBFS, fIN=1KHz.
- ^[3] For the supply voltage range, please refer to [DS981](#), [GW5AT series of FPGA Products Data Sheet](#) or [DS1103](#), [GW5A series of FPGA Products Data Sheet](#).

3 SARADC

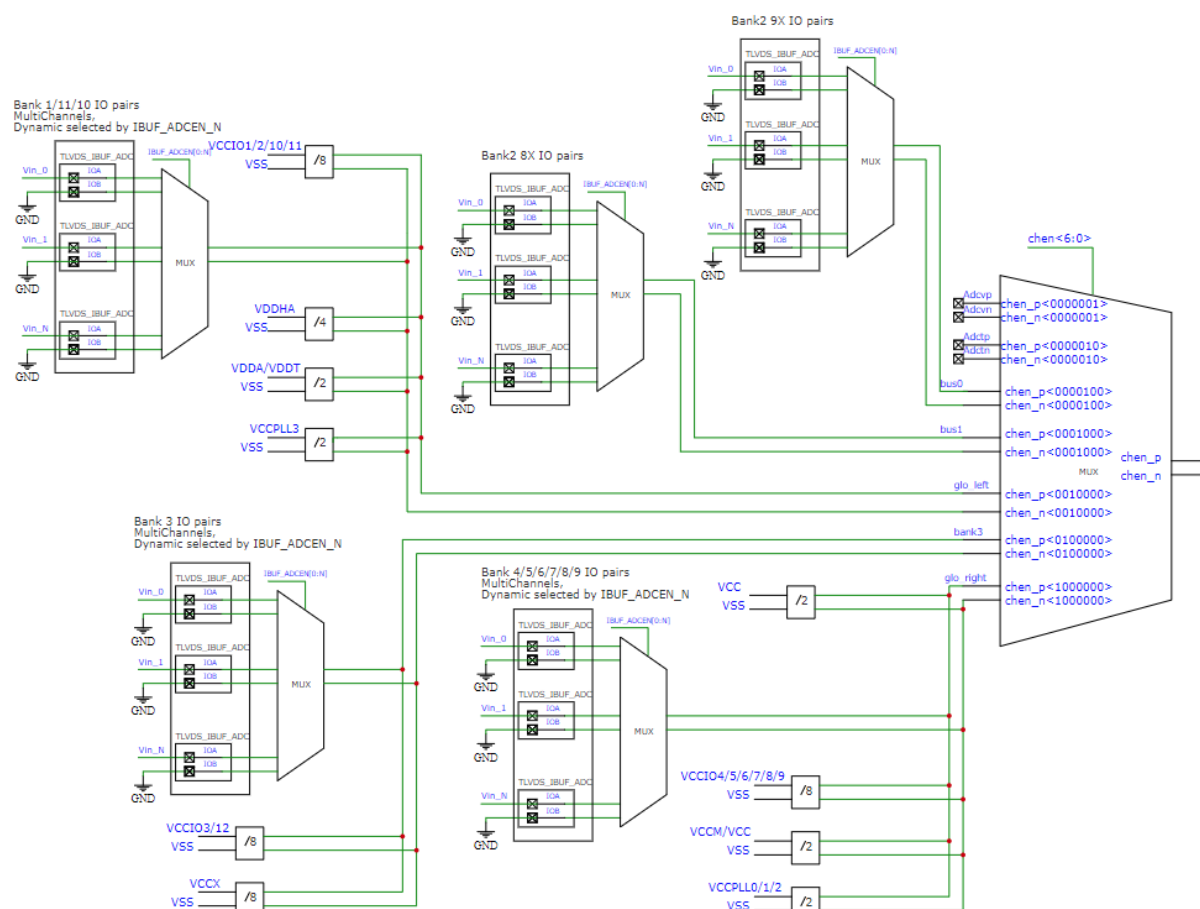
3.1 Devices Supported

Table 3-1 Supported Devices for SAR ADC

Product Family	Series	Device
Arora V	GW5A	GW5A-60
	GW5AT	GW5AT-60

3.2 Input Channel Selection (Voltage Divider Ratio)

The voltage-mode input channel selection block diagram of the GW5A-60/GW5AT-60 SAR ADC is as shown in [Figure 3-1](#).

Figure 3-1 SAR ADC Voltage-mode Input Channel Selection Block Diagram

The voltage-mode input channel selection voltage divider ratio table of the GW5A-60/ GW5AT-60 SAR ADC is as shown in [Table 3-2](#).

Table 3-2 Voltage-mode Input Channel Selection Voltage Divider Ratio Table

chen [6:0]	ADC Bus	Input Channel	Divider Ratio	BUF_EN[28:0]	Note
7'b0000001	adcv		1		
7'b0000010	adct		1		
7'b0000100	bus0		1		IOs on Bank 2
7'b0001000	bus1		1		IOs on Bank 2
7'b0010000	glo_left	VDD12M	4	BUF_EN[0]	
		VCCIO1	8	BUF_EN[1]	
		VCCIO2	8	BUF_EN[2]	
		VCCIO10	8	BUF_EN[15]	
		VCCIO11	8	BUF_EN[16]	
		VDDAM	2	BUF_EN[20]	
		VDDHAQ0	4	BUF_EN[22]	

chen [6:0]	ADC Bus	Input Channel	Divider Ratio	BUF_EN[28:0]	Note
7'b0010000	glo_left	VDDAQ0	2	BUF_EN[23]	
		VDDTQ0	2	BUF_EN[24]	
		VCCPLL3	2	BUF_EN[28]	
		PAD	1		BANK1/10/11 上 IO
7'b0100000	bank3	VCCIO3	8	BUF_EN[3]	
		VCCIO12	8	BUF_EN[4]	
		VCCX	8	BUF_EN[5]	
		PAD	1		IOs on Bank3
7'b1000000	glo_right	VCC	2	BUF_EN[7]	
		VCCLDO	8	BUF_EN[8]	
		VCCIO4	8	BUF_EN[9]	
		VCCIO5	8	BUF_EN[10]	
		VCCIO6	8	BUF_EN[11]	
		VCCIO7	8	BUF_EN[12]	
		VCCIO8	8	BUF_EN[13]	
		VCCIO9	8	BUF_EN[14]	
		VCCM	2	BUF_EN[17]	
		VCCPLL0	2	BUF_EN[25]	
		VCCPLL1	2	BUF_EN[26]	
		VCCPLL2	2	BUF_EN[27]	
		PAD	1		IOs on Bank 4/5/6/7/8/9

3.3 SAR ADC Instantiation

3.3.1 Port Diagram

Figure 3-2 SAR ADC Port Diagram (Static Input Channel)

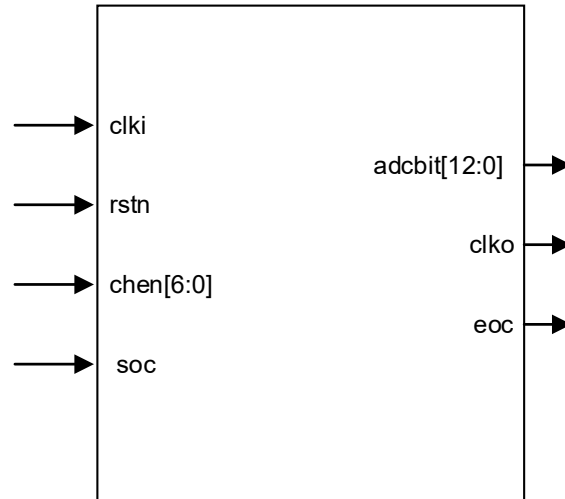
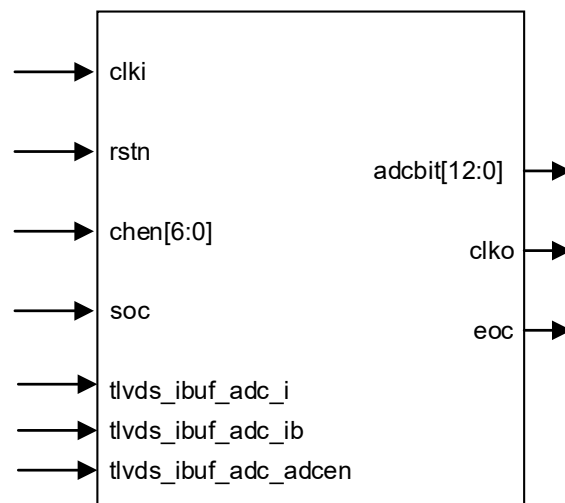


Figure 3-3 SAR ADC Port Diagram (Dynamic Input Channel)



3.3.2 Port Description

SAR ADC Port

Table 3-3 SAR ADC Port Description

Port	I/O	Description
chen[6:0]	input	input source selection bit [6:0] 7'b0000001: adcv 7'b0000010: adct 7'b0000100: bus0 7'b0001000: bus1 7'b0010000: glo_left 7'b0100000: bank3 7'b1000000: glo_right
clki	input	clk input
rstn	input	digital part reset signal, active low
soc	input	measurement request signal, valid rising edge
eoc	output	measurement completion signal, active high
adcbits[12:0]	output	bit[12:0] the measurement result output
clko	output	bypass clki

SAR ADC_IO_BUF Port

Table 3-4 SAR ADC_IO_BUF Description

Port	I/O	Description
tlvds_ibuf_adc_i	input	The input of the differential GPIO port A, which connects to the voltage to be measured.
tlvds_ibuf_adc_ib	input	The input of the differential GPIO port B, which connects to GND.
tlvds_ibuf_adc_adcen	input	The ADC enable signals (adcen) of differential GPIO channels. Only one ADC channel can be enabled at any given time. The combination of chen and adcen signals is used to achieve the desired dynamic control.

3.3.3 Parameter Description

Table 3-5 SAR ADC Configuration Interface Parameters

Parameter	Default Value	Description
Common	ADC/SAR ADC	SAR ADC
Enable Clock Division	Not selected	When selected, Clock Division Factor 1/2/3/4
Clock Source Selection	CLK	clk source

Parameter	Default Value	Description
Clock Source Selection	CLK	PLL or CLK, selecting 'PLL' restricts the clock source to clkout0 of PLL_T[1]. If 'CLK' is selected, an external user-supplied clock can be used instead.
BandGap Reference Voltage Selection	Internal	Internal or External 'Internal' selects the internal voltage reference, whereas 'External' requires an external reference (Requires dedicated pins Vrefp and Vrefn.).
Sample and Hold Sample Start Point	1	Fixed to 1
Sample and Hold Sample End Point	11	The value must be greater than or equal to the Sample and Hold Sample Start Point +10.
Digital to Analog Converter Sample Start Point	13	The value must be greater than or equal to the Sample and Hold Sample End Point +2.
Digital to Analog Converter Sample End Point	18	The value must be greater than or equal to the Digital to Analog Converter Sample Start Point +5
Counter Length	20	The value must be greater than or equal to the Digital to Analog Converter Sample End Point +2
Dynamic Input Source Enable (Voltage mode)	Not selected.	When selected, enables the tlvs_ibuf_adc ADC input interface (supports all bank inputs and multiple IO pairs).
glo_left (Voltage mode)	VDD12M(LV) VCC(EV)	When using IOs from Bank 1, 10, or 11—which correspond to power rails such as VDD12M(LV)/VCC(EV), VCCIO1_2_10_11, VDDA_MIPI, VDDHA_Q0, VDDA_Q0, VDDT_Q0, VCCPLL3, and the associated PADs—as ADC input channels, and the 'Dynamic Input Source Enable' option is not selected, the following physical constraint must be added: USE_ADC_SRC bus2 loc; loc: Specifies the location of the ADC input pin, for example IOR26 When 'Dynamic Input Source Enable' is selected, the tlvs_ibuf_adc ADC input interface is enabled. The A-side should be connected to the voltage to be measured, and the B-side must be connected to GND.
glo_right (Voltage mode)	VCC	VCC/VCC_LDO/VCCIO4_5_6_7_8_9/VCCM/VCCPLL0_1_2 / PAD

Parameter	Default Value	Description
glo_right (Voltage mode)	VCC	When using IOs from Bank 4, 5, 6, 7, 8, or 9—which correspond to PADS in these banks—as ADC input channels and the 'Dynamic Input Source Enable' option is not selected, the following physical constraint must be added: USE_ADC_SRC bus4 loc; loc: Specifies the location of the ADC input pin, for example IOR26 When 'Dynamic Input Source Enable' is selected, the tlvs_ibuf_adc ADC input interface is enabled. The A-side should be connected to the voltage to be measured, and the B-side must be connected to GND.
bank3 (Voltage mode)	VCCIO3	VCCIO3/VCCIO12/VCCX/PAD When using IOs from Bank 3—which correspond to power rails such as and the associated PADS—as ADC input channels, and the 'Dynamic Input Source Enable' option is not selected, the following physical constraint must be added: USE_ADC_SRC bus3 loc; loc: Specifies the location of the ADC input pin, for example IOR26 When 'Dynamic Input Source Enable' is selected, the tlvs_ibuf_adc ADC input interface is enabled. The A-side should be connected to the voltage to be measured, and the B-side must be connected to GND.
bus0	PAD	The 9 pairs of IOs in Bank 2, please refer to the pinout manual.
bus1	PAD	The 8 pairs of IOs in Bank 2, please refer to the pinout manual.

3.3.4 Instantiation Example:

Verilog Instantiation:

```
Gowin_ADC your_instance_name(
.adcbit(adcbt), //output [12:0] adcbt
.clko(clko), //output clko
.eoc(eoc), //output eoc
.clki(clki), //input clki
.chen(chen), //input [6:0] chen
.rstn(rstn), //input rstn
.soc(soc) //input soc
);
```

Vhdl Instantiation:

```
component Gowin_ADC
```

```
port (  
  adcbt: out std_logic_vector(12 downto 0);  
  clko: out std_logic;  
  eoc: out std_logic;  
  clki: in std_logic;  
  chen: in std_logic_vector(6 downto 0);  
  rstn: in std_logic;  
  soc: in std_logic  
);  
end component;  
your_instance_name: Gowin_ADC  
port map (  
  adcbt => adcbt,  
  clko => clko,  
  eoc => eoc,  
  clki => clki,  
  chen => chen,  
  rstn => rstn,  
  soc => soc  
);
```

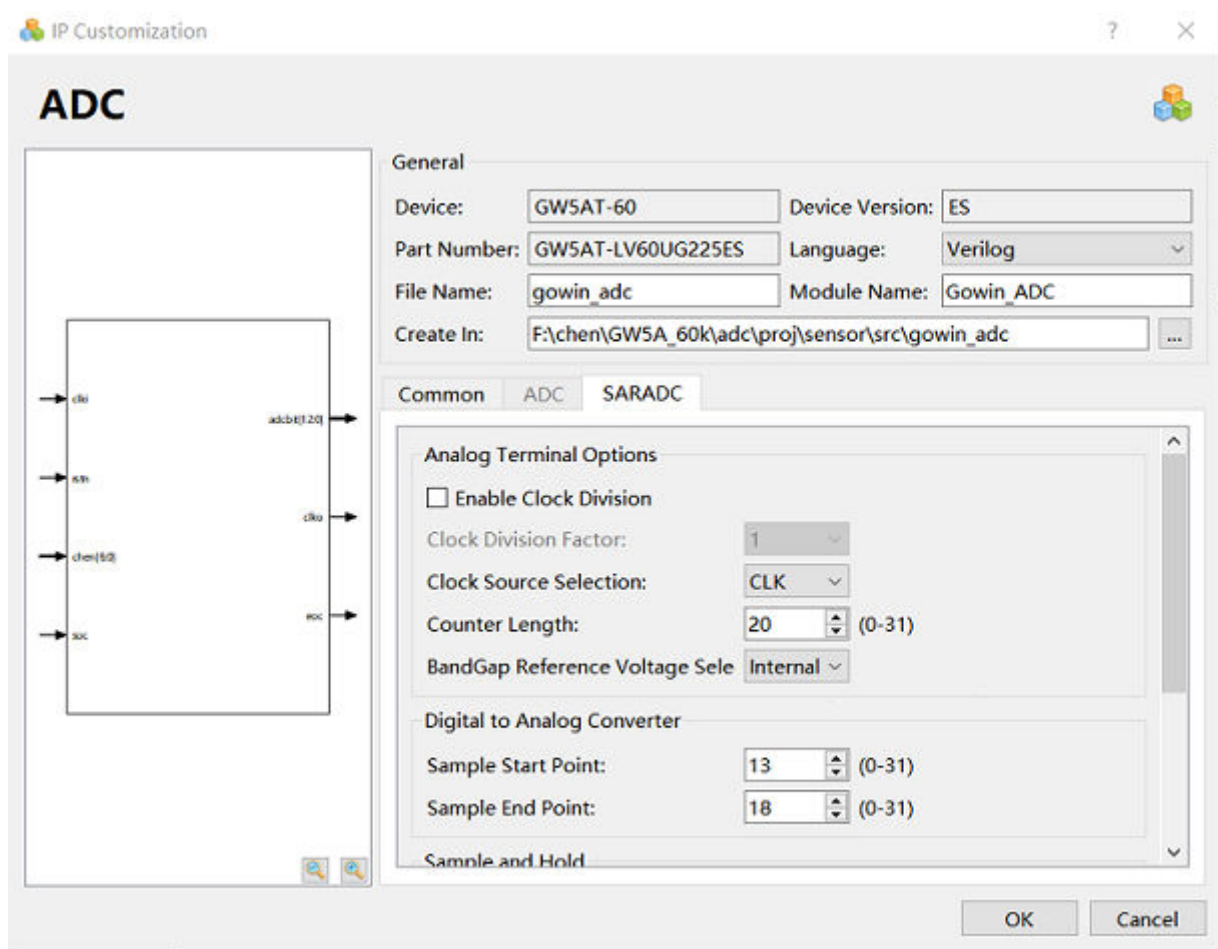
4 SAR ADC Configuration and Generation

Click “Tools > IP Core Generator” in Gowin Software to call and configure ADC.

4.1 SAR ADC Configuration

The configuration options for SAR ADC are shown in [Figure 4-1](#).

Figure 4-1 SAR ADC Configuration Page



4.2 Generation Files

After SAR ADC configuration, it will generate three files that are named after the "File Name". Take the default configuration as an example:

- "gowin_adc.v" file is a complete Verilog module to generate an instance of Gowin_ADC based on the user's IP configuration.
- "gowin_adc_tmp.v" is a template file for IP designs;
- "gowin_adc.ipc" file is an IP configuration file for users to load and configure the IP.

Note!

If VHDL is selected as the hardware description language, the first two files will be named with .vhd suffix.

